Basic Mobile App Development with Block-Based Programming

DICT Region V – Camarines Sur



What is Block-Based Coding?

Block-based coding, sometimes known as block based-programming, is coding within a programming language where instructions are mainly represented as blocks.

What is Block-Based Coding?

Block-based Programming uses a programming-primitive-as-puzzle-piece metaphor as a means of providing visual cues to the user as to how and where commands may be used.

How does it work?

Block-based programming provides visual clues through predefined groups of code (blocks), such as **events**, **controls**, **commands**, **loops**, **conditions**, and more, that can be selected, modified, and sequenced to create and run games, apps, and other programs.

Text-Based vs. Block-Based Programming



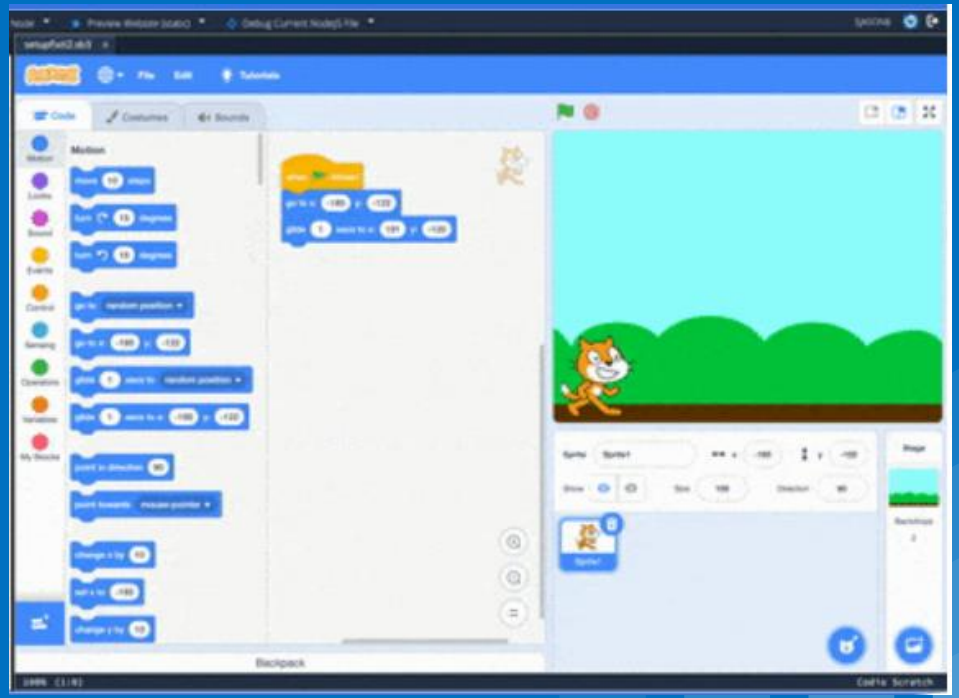
Scratch 3.0 and the same code in Javascript

How Do I Learn with Block-Based Coding?

There are several block-based coding platforms that can get you started learning to code. Here are some of our favorites

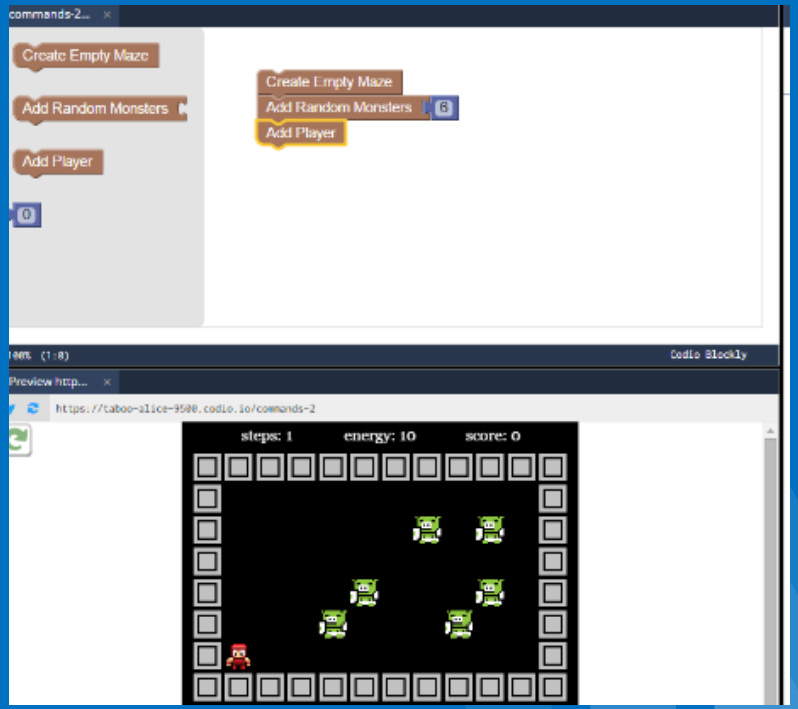
SCRATCH

Scratch is a popular block-based, visual coding environment that introduces programming concepts in an engaging way to students of all ages.



BLOCKLY

Blockly is another blocked-based programming tool that adds a visual programming environment and can be used in the Codio platform to help your courses to be more engaging and successful.



Block-Based Programming Tools

	Best for (Tool)	Type	Structure/Elements	Difficulty Level	Best for (Purpose)
	Almost all tools	Block	Motion Looks, Sound Events, Control, Sensing, Operators, Variables	Easy to Moderate	Building Simple Apps
	Micro:bit	Block	Basic Input, Music, LED, Radio, Loops, Logic, Math, Functions, Arrays, Text, Game Images, Pins, Serial, Control, Extensions	Moderate	Basic Robotics, Game Programming, Science Experiments
	Android Smartphones	Block	Control Logic, Match, Text, Lists, Dictionaries, Colours, Variables, Procedures	Moderate	Building Simple to Complex Apps

Other Block-Based Programming Apps

- Blockly, Strawbees CODE, SPIKE Prime

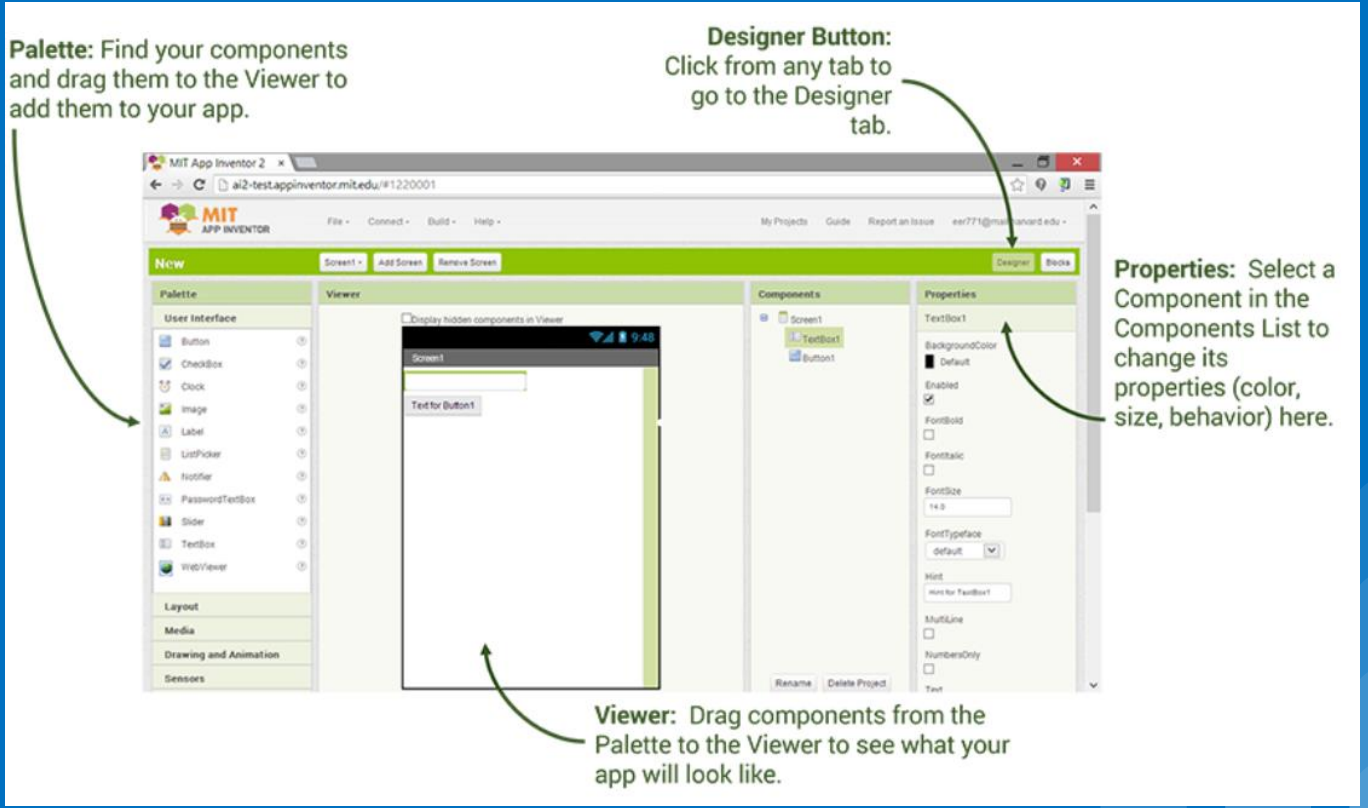
Getting Started with MIT App Inventor

App Inventor is a **cloud-based tool**, which means you can create apps for phones or tablets right in your web browser

App Inventor consists of the **Designer** and the **Blocks Editor.**

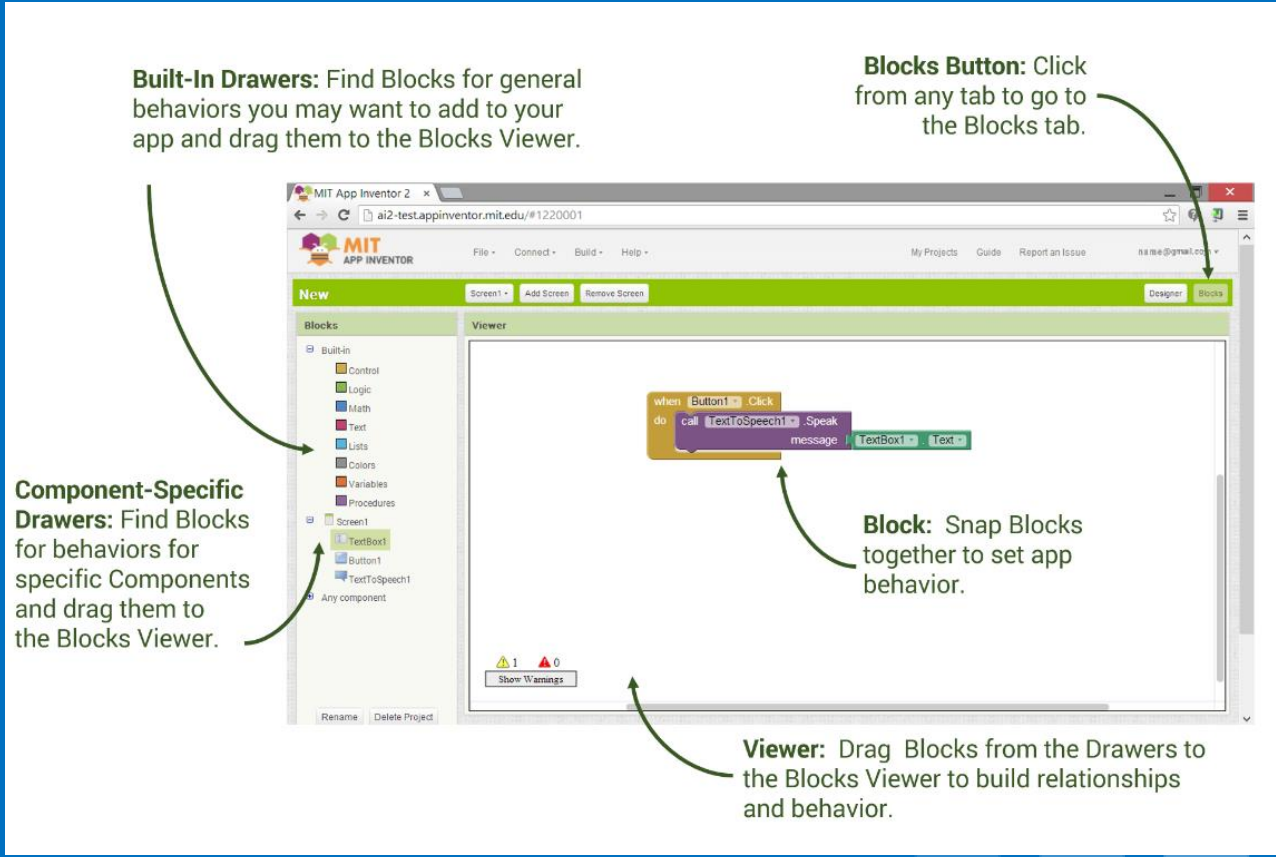
Designer and Blocks Editor

The **Designer** lets you create the app's interface:



Designer and Blocks Editor

The Blocks Editor lets you program the app's behavior by putting blocks together:





Let’s Build An App!

appinventor.mit.edu/